

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

December 2013

CL101 Fundamentals of Civil Engineering

Date: 02.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Enlist six branches of civil engineering and explain any two in detail. [07]

Q - 2 (a) Explain two fundamental principles of surveying with examples. [06]

(b) Attempt any Two. [08]

(i) An area of 49 cm^2 of a map represents an area of 2401 km^2 . Find scale and RF of a map.

(ii) How the chain can be adjusted if it is found too long and too short.

(iii) Describe indirect or reciprocal ranging.

Q - 3 (a) A 20m chain was found to be 8cm too long at the end of day's work after measuring 4000m. If the chain was correct before the commencement of the work, find the correct length of the line. [04]

(b) Attempt any Two. [10]

(i) Convert following WCB into Reduced Bearings: 49° , 240° , 133° , 335° , 190°

(ii) The following bearings were taken of a closed traverse ABCD. Calculate interior angles of the traverse.

Line	FB	BB
AB	$45^\circ 00'$	$225^\circ 00'$
BC	$123^\circ 30'$	$303^\circ 30'$
CD	$181^\circ 00'$	$1^\circ 00'$
DA	$289^\circ 00'$	$109^\circ 00'$

(iii) The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 5m. The position of the instrument was changed after 3rd, 7th and 10th readings. Draw out the form of a level book and enter the above readings properly. Assume the RL of the 1st point as 50.000. Calculate RL of all points and

apply usual checks.

SECTION - II

- Q - 4 i. Write full form of BOT. [01]
 ii. What is building byelaws? [03]
 iii. What are the sources of water? [03]
- Q - 5 (a) Prepare foundation plan for the plan shown in **Figure 1**. [07]
 (b) **Attempt any One.** [07]
 (i) Enlist the common building components in sequence from bottom to top with their nominal dimensions.
 (ii) Sketch typical wall section through door and window showing common building components.
- Q - 6 (a) Explain water conveyance system. [07]
 (b) **Attempt any One.** [07]
 (i) Sketch permanent way on embankment showing all the components of it.
 (ii) Discuss about traffic control devices.
 (iii) Explain water conservation with its importance.

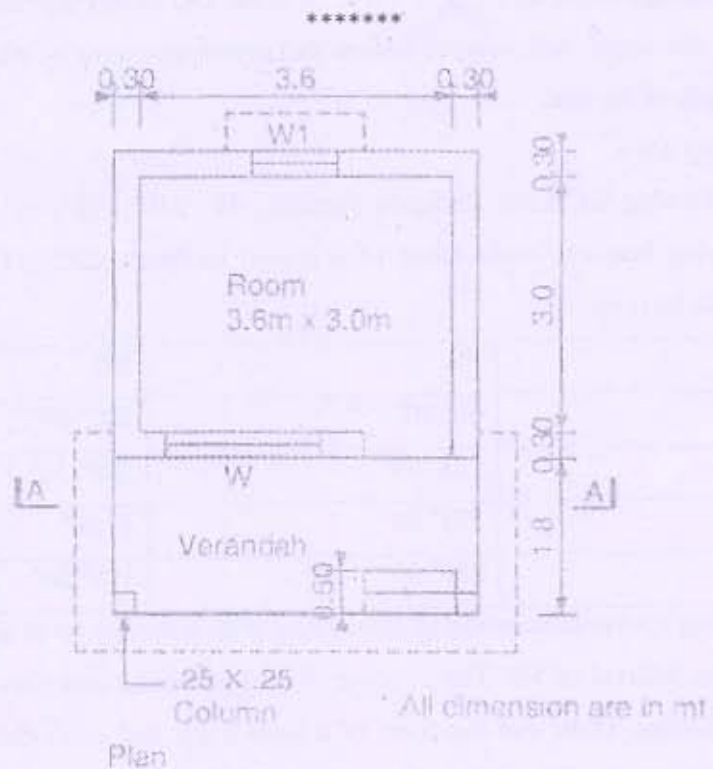


Figure 1